

11. Container with most water:

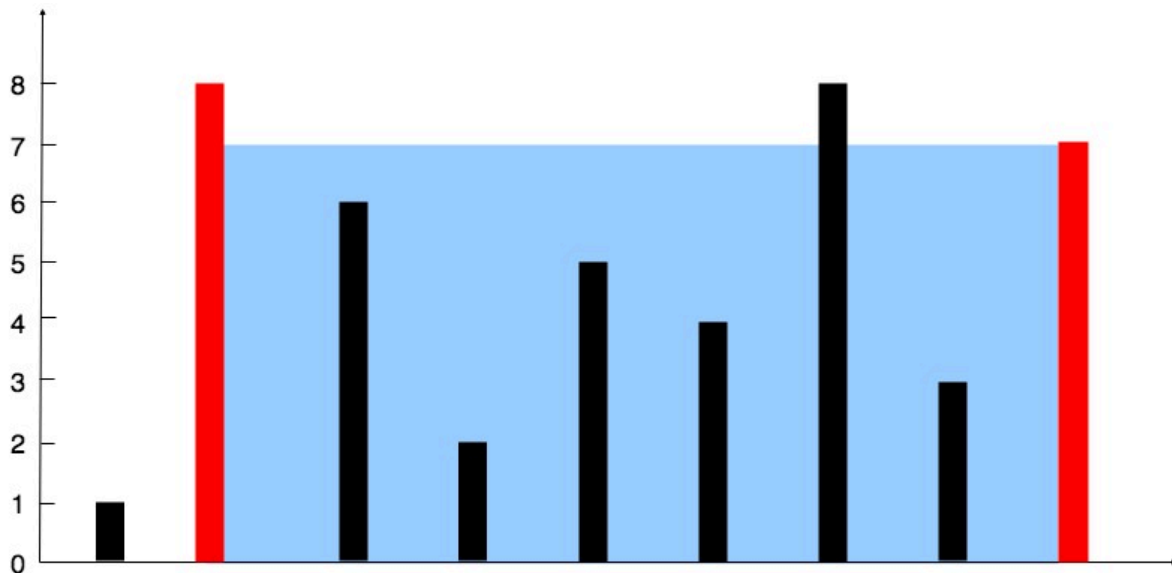
You are given an integer array `height` of length `n`. There are `n` vertical lines drawn such that the two endpoints of the `i`th line are `(i, 0)` and `(i, height[i])`.

Find two lines that together with the x-axis form a container, such that the container contains the most water.

Return *the maximum amount of water a container can store*.

Notice that you may not slant the container.

Example 1:



Input: `height = [1,8,6,2,5,4,8,3,7]`

Output: 49

Explanation: The above vertical lines are represented by array `[1,8,6,2,5,4,8,3,7]`. In this case, the max area of water (blue section) the container can contain is 49.

Example 2:

Input: height = [1,1]

Output: 1

Constraints:

- `n == height.length`
- `2 <= n <= 105`
- `0 <= height[i] <= 104`

Solution: time = $O(n)$ and space = $O(1)$

```
class Solution:
    def maxArea(self, height: List[int]) -> int:
        l = 0
        r = len(height)-1
        maxArea = 0
        while l <= r:

            width = r - l
            minHeight = min(height[l],height[r])
            area = minHeight * width
            maxArea = max(area,maxArea)
            if height[l] < height[r]:
                l += 1
            else:
                r -= 1
        return maxArea
```